Minseong Kim

AI Applications Engineer · LLM · ML for Operations

Cerritos, CA · medium2000@naver.com · minseong-kim.github.io/career/#ai

SUMMARY

Engineer applying AI to operations data, built on 15+ years of enterprise data systems. Recent independent projects use the Claude API for drafting and coherence checks, retrieval-augmented generation (RAG) for similar-document search, and Random Forest and Isolation Forest models for root-cause analysis and predictive maintenance on simulated manufacturing data. Background in safety-critical data validation for aviation (NTT DATA) and SQL performance tuning.

SKILLS

Languages	Java, C#, Python, TypeScript, JavaScript, SQL, Rust, Shell Script, Groovy
Back end & APIs	Spring Boot, Spring, ASP.NET / .NET Core, Node.js / Express, Django, FastAPI, RESTful API design, SOAP / JAX-WS, JSON / XML
Databases	Oracle, SQL Server, MySQL, PostgreSQL, MongoDB, Data modeling (ERwin), Query tuning
AI & ML	Claude API (LLM), Tool use, RAG, Random Forest, Isolation Forest, Anomaly detection, Predictive maintenance
Manufacturing & enterprise	MES, PLM, BOM / ECN, ERP & SAP integration, WMS, TMS, QMS, FDC / RCA
Automation	Selenium workflows, Web scraping, Scheduled & background jobs, Event-driven workflows, Data pipelines

INDEPENDENT AI PROJECTS

ClinicalMind – AI-Assisted Clinical Study Design — Three Claude API workflows (draft generation, real-time CDISC coherence check, RAG search of similar protocols) and an immutable audit trail designed for 21 CFR Part 11. Angular 19 architecture patterns.

MES Process Innovation Console (Solar) — Ingot-to-module MES simulation with AI-assisted root-cause analysis and predictive maintenance using Random Forest and Isolation Forest models.

Battery MES·AI Integration Platform — Six-system cascade (MES, FDC, EMS, RMS, QMS, MCS) with defect classification, anomaly detection and battery-life models; Spring Boot REST APIs and LLM diagnostic explanations.

Switchgear MES·AI Bridge — Bilingual console modeling FDC root-cause analysis, automated HiPot failure response, QMS re-inspection and HQ ↔ plant ECN/BOM coordination.

EXPERIENCE

Independent Technical Projects

08/2025 – Present · Cerritos, CA

MES / AI portfolio (self-directed, non-commercial)

- Built four live demos covering MES/AI integration and LLM-assisted clinical study design (see Independent AI Projects).
- Implemented Claude API workflows with RAG retrieval and real-time validation, keeping humans in the decision loop.
- Modeled anomaly detection and predictive maintenance with Random Forest and Isolation Forest on simulated plant data.

FNS Inc

11/2021 – 08/2025 · Torrance, CA

IT Lead Manager (Full-Stack Developer) · global logistics

- Tuned and validated SQL Server, Oracle and MySQL databases.

Nmsi Inc

06/2020 – 10/2021 · Brea, CA

IT Lead Manager (Full-Stack Developer) · mortgage lending

- Designed, built and maintained secure, service-oriented mortgage applications for origination, underwriting and servicing on ASP.NET, .NET Core, C# and SQL Server.

Very J 11/2019 – 06/2020 · Los Angeles, CA

IT Manager (Full-Stack Developer) · retail and warehouse

- Built a real-time inventory synchronization system that transmitted inventory data between warehouses and retail stores automatically.

Daegu Bank 05/2016 – 12/2016 · Korea

Project Leader · next-generation banking system

- Designed and developed core banking applications and integrated financial services; led UAT and production implementation.

PV Technology 12/2015 – 05/2016 · Korea

Project Leader · Warehouse Management System

- Led a WMS covering receiving, storage, picking, packing and shipping, including integration, UAT and user training.

Korea Post Information Center 12/2013 – 06/2015 · Korea

Project Leader · Smart Electronic Banking

- Led development of mobile and online banking applications on an integrated mobile banking platform.

Samsung Electronics 06/2011 – 12/2013 · Gyeonggi-do, Korea

Project Leader (Software Engineer) · PLM and System LSI

- Led the System LSI IP management system for cataloging and reusing semiconductor design IP through tree browsing and parametric search.
- Built a technical document workspace, related-object search and usage analytics (most-downloaded documents, top users).

LG Hitachi — *Software Engineer · financial services* 02/2011 – 06/2011 · Korea

Nexon Mobile — *Software Engineer · mobile web* 09/2010 – 01/2011 · Korea

NTT DATA 01/2007 – 09/2010 · Fukuoka, Japan

Software Engineer · aeronautical information system

- Validated flight routes, navigation aids and airspace data on the 28-day AIRAC cycle, with logic that caught errors in safety-critical data.
- Improved performance on large spatial and geographic data sets through code review and tuning.

Neobill — *Software Engineer · electronic billing and payments* 03/2006 – 01/2007 · Korea

Moazio — *Software Engineer · mobile games* 09/2005 – 03/2006 · Korea

EDUCATION

Georgia Institute of Technology 2027

M.S. Computer Science ADMITTED

Long Beach City College 2026 – now

A.S. Electrical Technology – Automation Technician

IN PROGRESS

California University of Management and Sciences 2019

M.S. Computer Information Systems

Jeonju University, Korea 2006

B.E. Information and Communication Engineering · GPA 3.99 / 4.5

CERTIFICATIONS

Oracle Certified Professional (OCP), Database 10g 2010

Sun Certified Business Component Developer (SCBCD) 2010

Sun Certified Web Component Developer (SCWCD) 2010

Sun Certified Java Programmer (SCJP) 2010

JLPT N1 (Japanese, highest level) 2014

Languages

Korean (native) · English (professional) · Japanese (business, JLPT N1)

Military service

Republic of Korea Air Force, full term completed